

CODING MASTERS:
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SEM 1 PROJECT

‘LOSER???’

hAmLeY's

piano



Our inspiration for our first sem final project was the floor piano we usually see at Hamley's. As children, we were never allowed on it because of the shop employees, so we made our own version of it in ODT

IDEATIONS



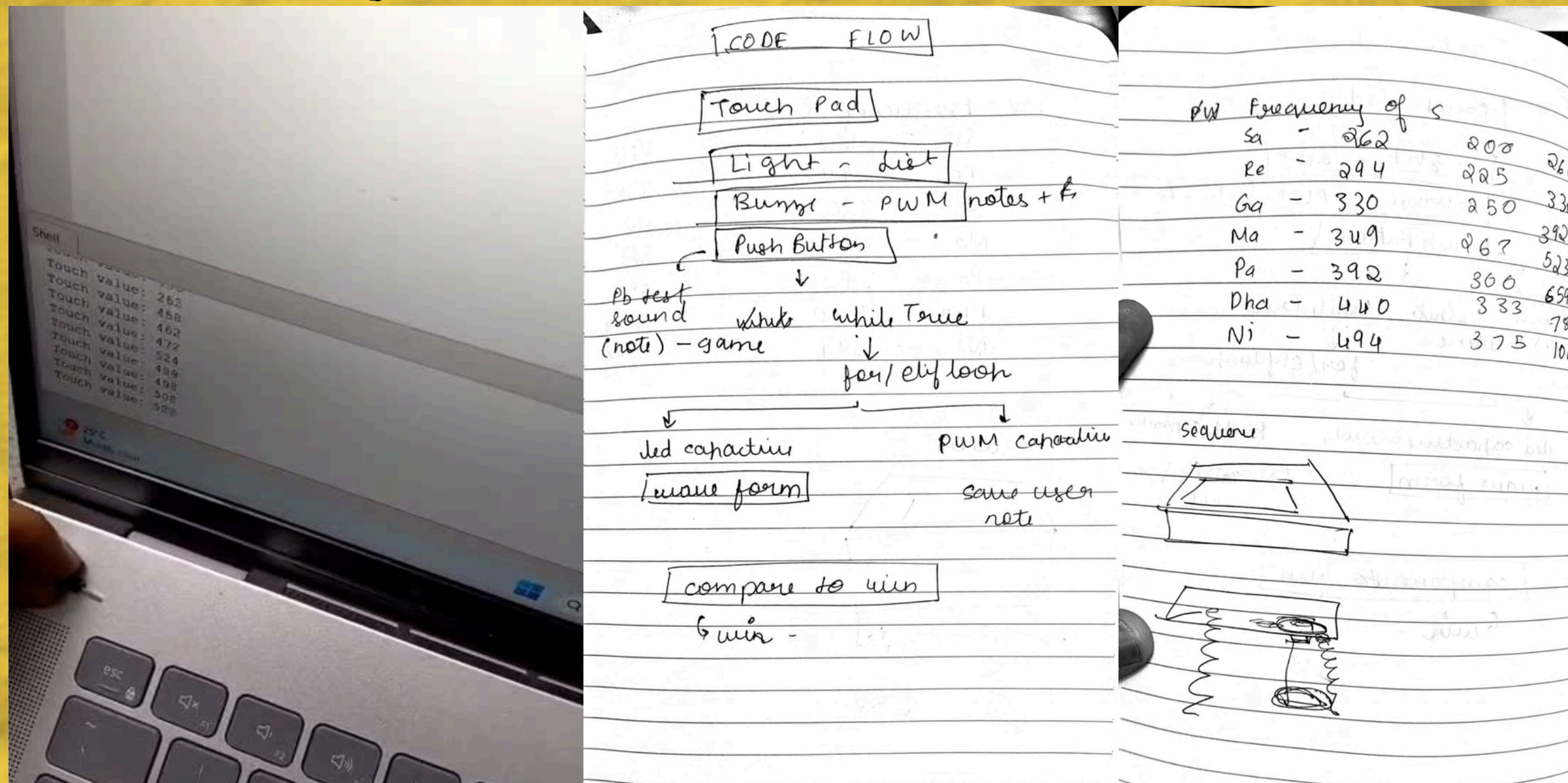
We ideated on how to make our piano different to the hamley's piano, *and* make it challenging, since we clearly aren't the same puny kids as earlier.

INITIAL TESTING



We split the required codes. Naviya focused on codes for capacitive touch, led patterns, and arcade PWM music. Manya focused on PWM codes for different intonations, pattern for the same, and the user logic.

CODING LOGIC



The collage consists of three parts:

- Laptop Screen:** Displays a terminal window with the following text:


```

      Touch value: 252
      Touch value: 252
      Touch value: 458
      Touch value: 462
      Touch value: 472
      Touch value: 524
      Touch value: 489
      Touch value: 498
      Touch value: 508
      Touch value: 522
      
```
- Handwritten Code Flowchart:**

```

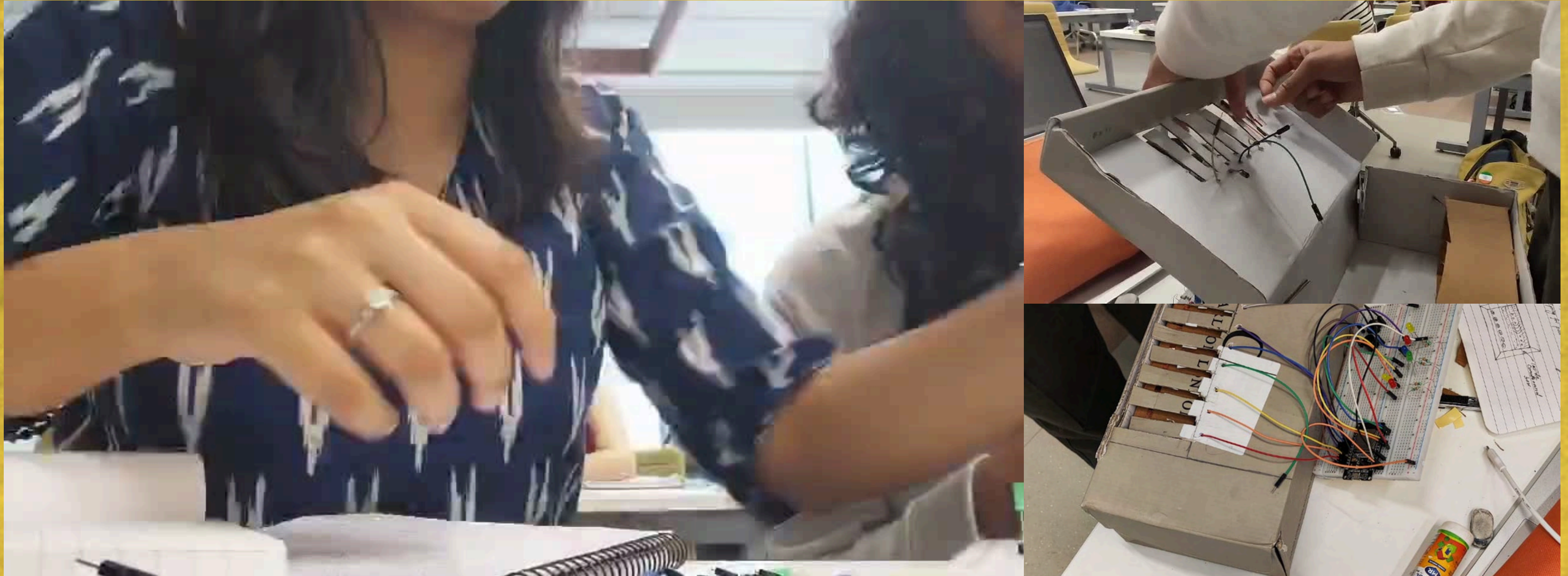
      graph TD
      subgraph "CODE FLOW"
      TP[Touch Pad] --> LD[Light - dist]
      LD --> B[PWM notes + f]
      B --> PB[Push Button]
      end
      PB --> PB_test[Pb test]
      PB_test --> sound[sound]
      PB_test --> game["(note) - game"]
      PB_test --> while_true["while True"]
      while_true --> for_loop["for/elif loop"]
      for_loop --> led_capacitor["led capacitor"]
      for_loop --> pwm_capacitor["pwm capacitor"]
      led_capacitor --> image_form[image form]
      pwm_capacitor --> same_user_note[same user note]
      image_form --> compare_win[compare to win]
      compare_win --> guess[guess -]
      
```
- Handwritten Table of Musical Note Frequencies:**

Note	Frequency (Hz)	Frequency (Hz)	Frequency (Hz)
Sa	262	200	262
Re	294	225	294
Ga	330	250	330
Ma	349	267	349
Pa	392	300	392
Dha	440	333	440
Ni	494	375	494

We defined our coding logic, and checked the feasibility of each code with constant checks at each step.

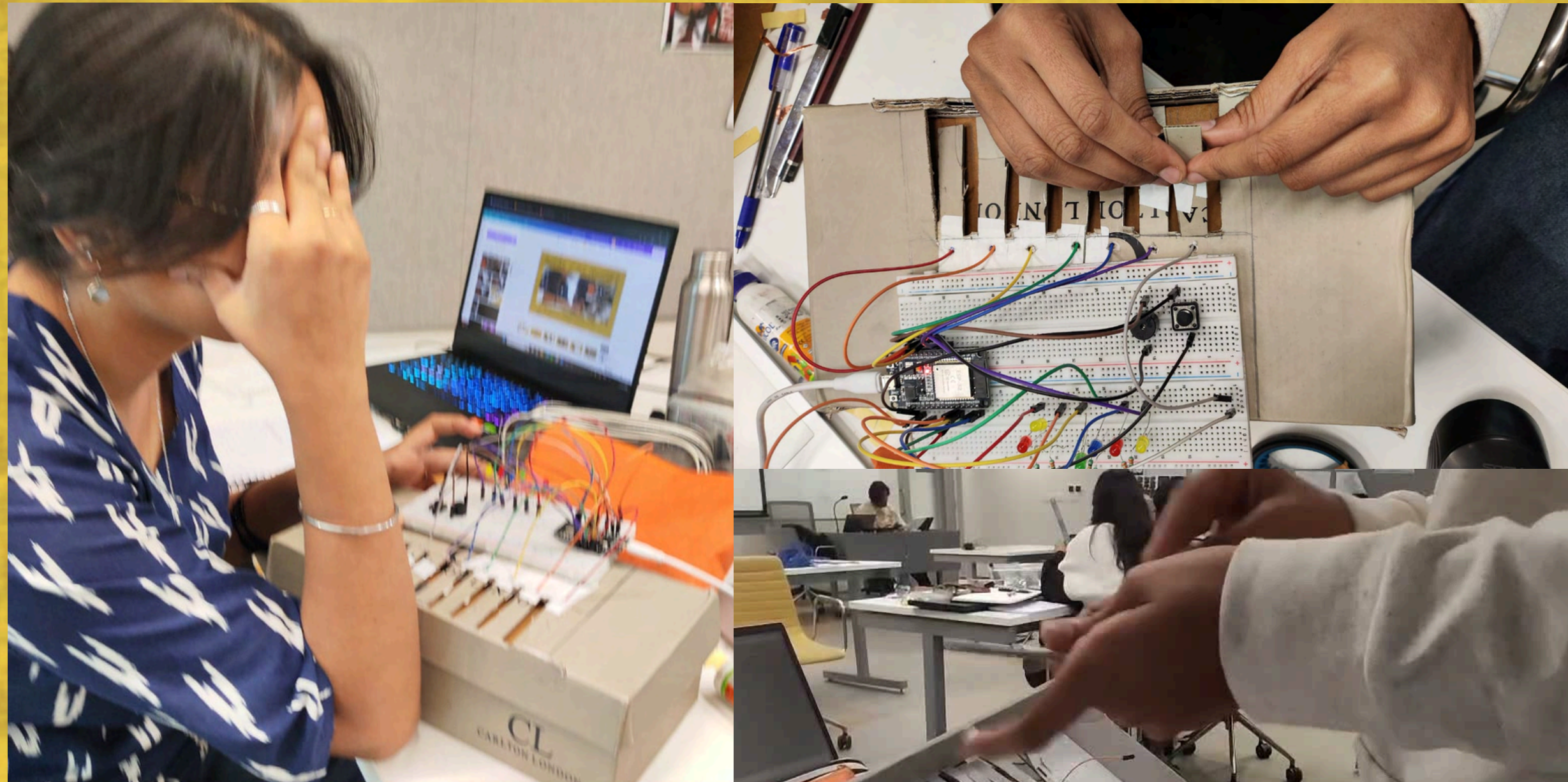
We defined frequencies, and noted important values to ensure no later confusion.

PUTTING IT ALL TOGETHER



The final assembly was once again split up to ensure efficiency. Naviya worked on the circuit board, documentation, and final polish of the code, while Manya made the working piano, and finalised connections.

PAIN POINTS



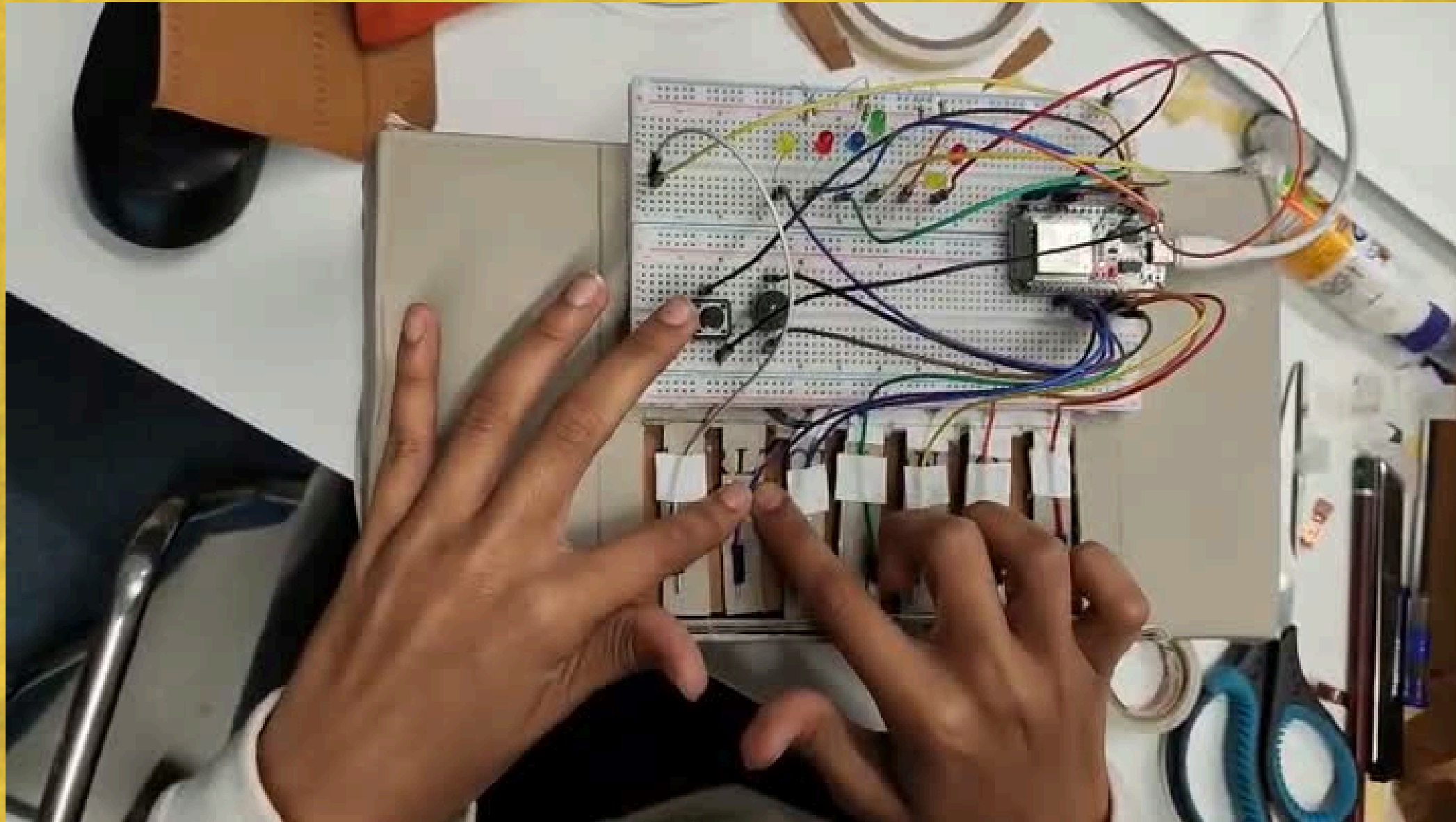
We faced problems in the following points:

Code logic to give enough time to the user to play the game.

Final outer covering wasn't sturdy enough for push gaming

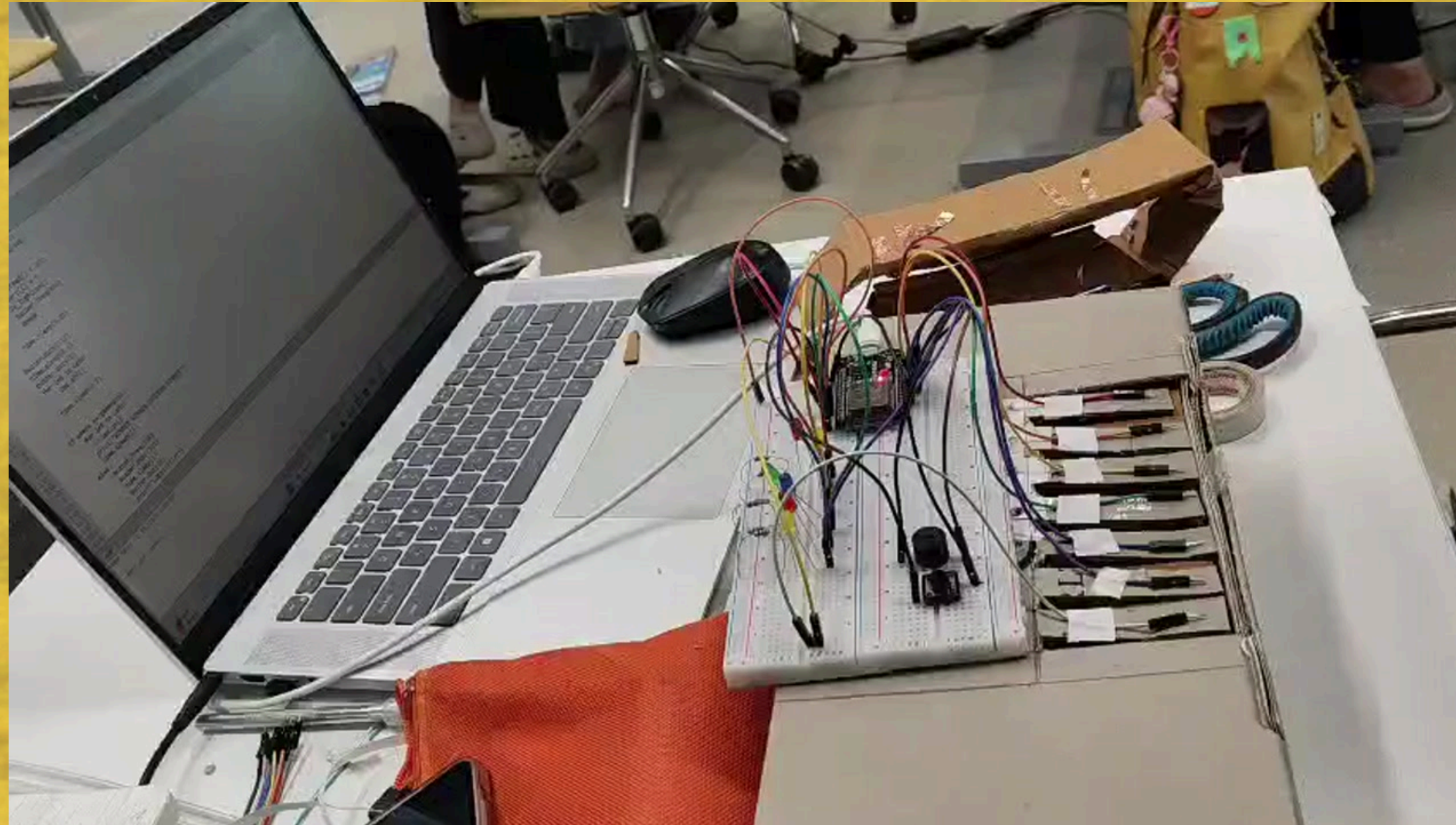
Breadboard couldn't be integrated into the physical product in an attractive manner.

REWORKING



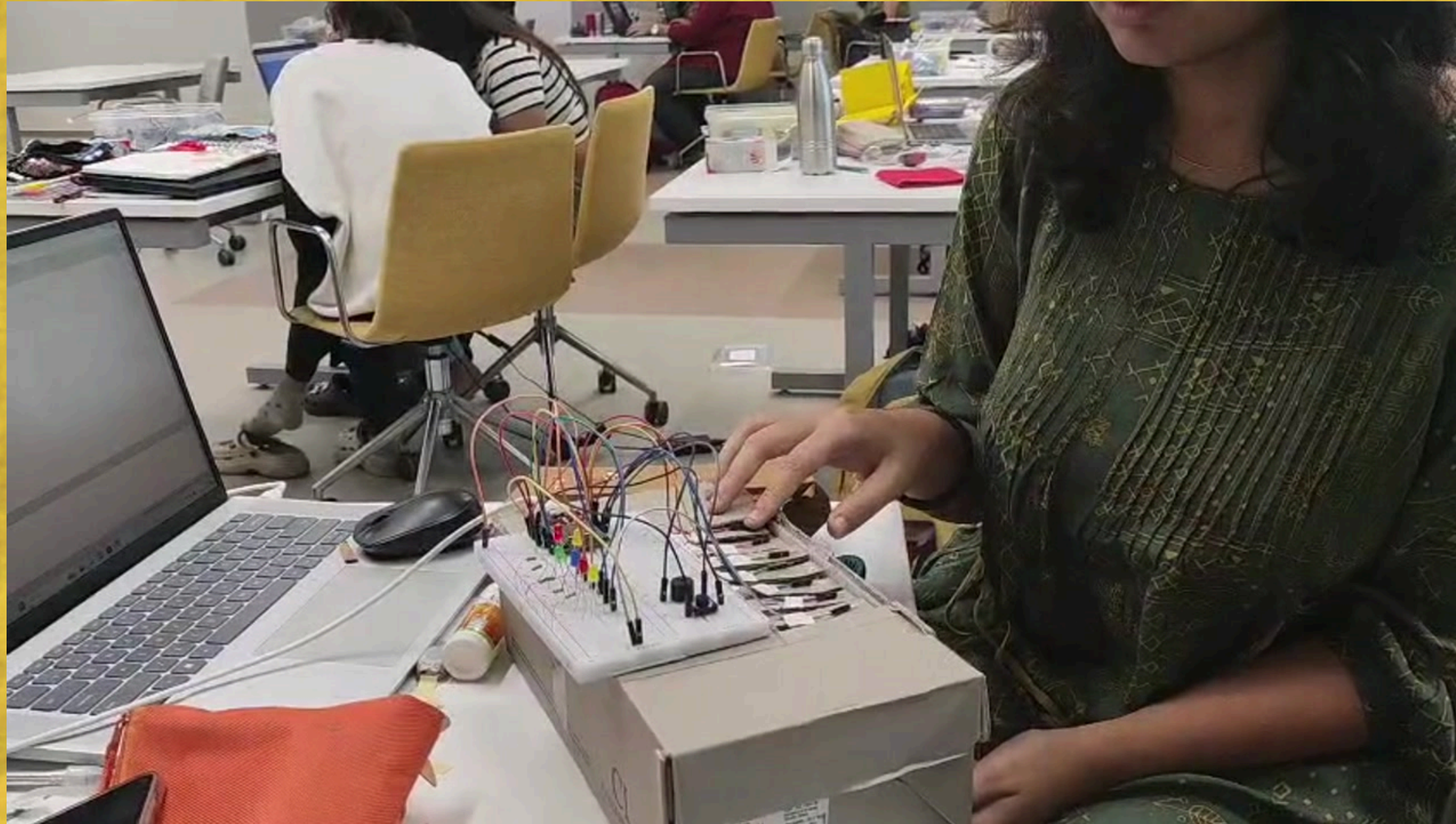
We went back to the drawing board, and ideated on how to work through our problems. We restructured our physical product to accomodate our problems, and executed the final project.

FINAL PRODUCT



A capacitive, memory-based game, that *no one* can win!

USER TESTING



We really do mean no one.

CONTRIBUTIONS

Naviya:

- Capacitive touch code
- Led Pattern code
- Arcade game tune
- Breadboard connections
- Documentation

Manya:

- PWM code
- Tune coding
- User logic section
- Physical Piano Product

